

On the Lagrange Inequality

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1 Introduction

One of the most fundamental theorems in calculus states that, under suitable conditions on a real valued function of a real variable, the average rate of increase of the function on any closed bounded interval is a value assumed by the derivative of the function at an interior point of that interval. The result is known as

Theorem 1.1 (The classical mean value theorem: MVT)

Let $f : [a, b] \rightarrow \mathbb{R}$ be such that

- (i) f is continuous on $[a, b]$;
- (ii) f is differentiable in (a, b) .

Then there is a point $c \in (a, b)$ such that $f(b) - f(a) = f'(c)(b - a)$.

Remark 1.2 The above theorem cannot be generalised to the case when the range of f is contained in a space of dimension at least two. In fact, consider the function $f : \mathbb{R} \rightarrow \mathbb{R}^2$ defined by $f(t) = (\sin t, \cos t)$ for any $t \in \mathbb{R}$. Fix $a, b \in \mathbb{R}$ with $a < b$. Suppose that for some $c \in (a, b)$, we have $f(b) - f(a) = f'(c)(b - a)$. Then

$$(\sin b - \sin a, \cos b - \cos a) = (b - a)(\cos c, -\sin c) \Rightarrow$$

$$\sin b - \sin a = (b - a) \cos c \quad \text{and} \quad \cos b - \cos a = -(b - a) \sin c.$$

Now taking the square on both sides of the two equations and summing we get

$$2 - 2(\cos b \cos a + \sin b \sin a) = (b - a)^2 \Rightarrow \left| \sin \left(\frac{b - a}{2} \right) \right| = \left| \frac{b - a}{2} \right|$$

where the last equation is obtained using the half-angle formula. This is a contradiction because $|\sin \theta| < |\theta|$ for any $\theta \in \mathbb{R}$ with $\theta \neq 0$.

However, in the above counter example we have the following estimate

$$|f(b) - f(a)| \leq |b - a| = |b - a| \sup_{c \in (a, b)} |f'(c)|.$$

We will generalise this estimate in the following theorem.

Theorem 1.3 Let $(V, \|\cdot\|_V)$ be a normed space and let $a, b \in \mathbb{R}$ with $a < b$. Let $f : [a, b] \rightarrow V$ be a continuous function. If f is differentiable in (a, b) , then there exists $\xi \in (a, b)$ such that

$$\|f(b) - f(a)\|_V \leq (b - a) \|f'(\xi)\|_V.$$

Proof: Consider the function $\phi : [a, b] \rightarrow \mathbb{R}$ defined by

$$\phi(t) = (b - a) \|f(t) - f(a)\|_V - (t - a) \|f(b) - f(a)\|_V \quad \text{for any } t \in [a, b].$$

ϕ is continuous on $[a, b]$ and $\phi(a) = \phi(b)$. Hence, ϕ achieves its minimum or maximum value at some point $\xi \in (a, b)$.

If ϕ achieves its minimum value at ξ , then for any $\eta > \xi$, we have $\phi(\xi) \leq \phi(\eta)$. That is,

$$(b - a) \|f(\xi) - f(a)\|_V - (\xi - a) \|f(b) - f(a)\|_V \leq (b - a) \|f(\eta) - f(a)\|_V - (\eta - a) \|f(b) - f(a)\|_V.$$

Thus, $(\eta - \xi) \|f(b) - f(a)\|_V \leq (b - a) [\|f(\eta) - f(a)\|_V - \|f(\xi) - f(a)\|_V]$. Therefore

$$\|f(b) - f(a)\|_V \leq (b - a) \left\| \frac{f(\eta) - f(\xi)}{\eta - \xi} \right\|_V \Rightarrow \|f(b) - f(a)\|_V \leq (b - a) \lim_{\eta \rightarrow \xi^+} \left\| \frac{f(\eta) - f(\xi)}{\eta - \xi} \right\|_V$$

which gives $\|f(b) - f(a)\|_V \leq (b - a) \|f'(\xi)\|_V$.

If ϕ achieves its maximum value at ξ , then for any $\eta < \xi$, we have $\phi(\xi) \leq \phi(\eta)$. That is,

$$(b - a) \|f(\xi) - f(a)\|_V - (\xi - a) \|f(b) - f(a)\|_V \geq (b - a) \|f(\eta) - f(a)\|_V - (\eta - a) \|f(b) - f(a)\|_V.$$

Thus, $(\eta - \xi) \|f(b) - f(a)\|_V \geq (b - a) [\|f(\eta) - f(a)\|_V - \|f(\xi) - f(a)\|_V]$. Therefore

$$\|f(b) - f(a)\|_V \leq (b - a) \left\| \frac{f(\eta) - f(\xi)}{\eta - \xi} \right\|_V \Rightarrow \|f(b) - f(a)\|_V \leq (b - a) \lim_{\eta \rightarrow \xi^-} \left\| \frac{f(\eta) - f(\xi)}{\eta - \xi} \right\|_V$$

which gives $\|f(b) - f(a)\|_V \leq (b - a) \|f'(\xi)\|_V$. $\square$

Exercise 1.4 Let $(V, \|\cdot\|_V)$ be a normed space and let $a, b \in \mathbb{R}$ with $a < b$. Let $f : [a, b] \rightarrow V$ and $g : [a, b] \rightarrow \mathbb{R}$ be two continuous functions. Assume f and g are differentiable in (a, b) with $g'(t) > 0$ for any $t \in (a, b)$. Prove that there exists $\xi \in (a, b)$ such that

$$\frac{\|f(b) - f(a)\|_V}{g(b) - g(a)} \leq \frac{\|f'(\xi)\|_V}{g'(\xi)}.$$

Hint: Consider the function $\phi : [a, b] \rightarrow \mathbb{R}$ defined by

$$\phi(t) := (g(b) - g(a)) \|f(t) - f(a)\|_V - (g(t) - g(a)) \|f(b) - f(a)\|_V \quad \text{for any } t \in [a, b].$$

Our next result is on the functions with zero Fréchet derivative.

Theorem 1.5 Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be two normed spaces. Let Ω be an open connected subset of V and $f : \Omega \rightarrow W$ be a differentiable function such that $Df(x) = 0$ for any $x \in \Omega$. Then f is a constant function.

Proof: Let us fix $x_0 \in \Omega$ and set

$$\Omega_1 := \{x \in \Omega: f(x) = f(x_0)\} \quad \text{and} \quad \Omega_2 := \{x \in \Omega: f(x) \neq f(x_0)\}.$$

Then $\Omega = \Omega_1 \cup \Omega_2$.

Now since, f is differentiable, then f is continuous and hence, Ω_2 is an open.

Now let us prove that Ω_1 is also open. Let $y \in \Omega_1$.

$$y \in \Omega \text{ open} \Rightarrow \exists \delta > 0 \text{ such that } B(y, \delta) \subset \Omega.$$

Let us prove that $B(y, \delta) \subset \Omega_1$. That is, let us prove $f(x) = f(x_0)$ for any $x \in B(y, \delta)$.

Let $x \in B(y, \delta)$. If $x = y$ then $x \in \Omega_1$. Now assume that $x \neq y$ and let us consider the function $\varphi : [0, 1] \rightarrow W$ defined by

$$\varphi(t) := f(y + t(x - y)) \quad \forall t \in [0, 1].$$

This function is well defined because, for any $t \in [0, 1]$, $y + t(x - y) \in B(y, \delta) \subset \Omega$.

Notice that $\varphi(0) = f(y)$, $\varphi(1) = f(x)$ and by Chain Rule $\varphi'(t) = Df(y + t(x - y))(x - y) = 0$. Hence, the function $\varphi : [0, 1] \rightarrow W$ has zero derivative in $[0, 1]$. From Lagrange inequality, it follows that there exists $\xi \in (0, 1)$ such that

$$\|\varphi(1) - \varphi(0)\|_W \leq (1 - 0) \|\varphi'(\xi)\|_W = 0.$$

Hence, $\varphi(1) = \varphi(0)$; that is, $f(x) = f(y) = f(x_0)$.

We then have

$$\Omega = \Omega_1 \cup \Omega_2, \quad \Omega_1 \cap \Omega_2 = \emptyset, \quad \Omega_1, \Omega_2 \text{ open}, \quad \Omega_1 \neq \emptyset.$$

Therefore, since Ω is connected, it follows that $\Omega_2 = \emptyset$. That is, $\Omega = \Omega_1$ and hence, $f(x) = f(x_0)$ for any $x \in \Omega$. So f is a constant function. $\square$

In finite dimension, Theorem 1.5 reads as follows

Theorem 1.6 *Let Ω be an open connected subset of $\mathbb{R}^n$. If $f : \Omega \rightarrow \mathbb{R}$ is a function such that $\nabla f(x) = 0$ for any $x \in \Omega$, then f is a constant function.*

Exercise 1.7

1. Prove that $\arctan(\frac{y}{x}) + \arctan(\frac{x}{y}) = \frac{\pi}{2}$ for any $x > 0, y > 0$.
2. Prove that $\arctan(\frac{y}{x}) + \arctan(\frac{x}{y}) = \frac{\pi}{2}$ for any $x > 0, y < 0$.
3. Using the function $f(x, y) := \arctan(\frac{y}{x}) + \arctan(\frac{x}{y})$ defined in $\mathbb{R}^2$ except the axes, what can you say about Theorem 1.5 when Ω is not connected?

Exercise 1.8 Let $(V_1, \|\cdot\|_{V_1})$ and $(V_2, \|\cdot\|_{V_2})$ be two normed spaces. Let O be a convex open subset of V_1 and $f : O \rightarrow V_2$ be a differentiable function. If there exists $c \geq 0$ such that $\|Df(x)\|_{\mathcal{L}(V_1, V_2)} \leq c$ for any $x \in O$, then prove that f is a Lipschitz function with Lipschitz constant less or equal c .

The next lecture will be on convex sets and convex functions.